Team Member: Ashlee Zamora-Alvarez, Carson Eberhardt, Chase Rager, Nick Hajduch

CIVE 202-150

Project #2

Python Code	Annotations
`import pandas as pd` `import numpy as np`	This section imports the pandas and numpy libraries into the program. These libraries allow the program to handle numerical calculations and structured data if needed.
`UNIT_WEIGHT_WATER = 62.4` `CUBIC_YARD_FT3 = 27`	These lines define constant values used throughout the mix design calculations. The unit weight of water is set to 62.4 lb per cubic foot, and one cubic yard is defined as 27 cubic feet.

`def calculate_water_weight_Q(cement_A, fly_ash_B, silica_fume_C, other_scm_D, wc_ratio_E):`
`total_cementitious = cement_A + fly_ash_B + silica_fume_C + other_scm_D`
`water_weight_Q = total_cementitious * wc_ratio_E return water_weight_Q`
	This function calculates the water weight by first adding all cementitious materials together and then multiplying by the water-cement ratio. It returns the required water weight for the mix.

`def calculate_volume_R(cement_A, sg_cement_J):` `return cement_A / (sg_cement_J * UNIT_WEIGHT_WATER)`	This function calculates the volume of cement by dividing the cement weight by its specific gravity times the unit weight of water.
`def calculate_volume_S(fly_ash_B, sg_fly_ash_K):` `return fly_ash_B / (sg_fly_ash_K * UNIT_WEIGHT_WATER)`	This function calculates the volume of fly ash using the same formula structure as the cement volume calculation.
`def calculate_volume_T(silica_fume_C, sg_silica_fume_L): return silica_fume_C / (sg_silica_fume_L * UNIT_WEIGHT_WATER)`	This function calculates the volume of silica fume based on its weight and specific gravity.
`def calculate_volume_U(other_scm_D, sg_other_scm_M): return other_scm_D / (sg_other_scm_M * UNIT_WEIGHT_WATER)`	This function calculates the volume of other supplementary cementitious materials using the same engineering formula.
`def calculate_air_volume_V(air_content_F):` `return (air_content_F / 100) * CUBIC_YARD_FT3`	This function converts air content percentage into an absolute air volume in cubic feet.
`def calculate_water_volume_W(water_weight_Q):` `return water_weight_Q / UNIT_WEIGHT_WATER`	This function converts water weight into water volume by dividing by the unit weight of water.
`def calculate_total_aggregate_volume_X(R, S, T, U, V, W):` `return CUBIC_YARD_FT3 - (R + S + T + U + V + W)`	This function calculates the total aggregate volume by subtracting all other material volumes from one cubic yard.

`def calculate_fine_aggregate_Y(percent_fine_G, sg_fine_N, volume_X): return volume_X * (percent_fine_G / 100) * sg_fine_N * UNIT_WEIGHT_WATER`	This function calculates the fine aggregate weight using the percentage of fine aggregate, its specific gravity, and the total aggregate volume.
`def calculate_coarse_aggregate_Z(percent_coarse_H, sg_coarse_O, volume_X): return volume_X * (percent_coarse_H / 100) * sg_coarse_O * UNIT_WEIGHT_WATER`	This function calculates the coarse aggregate weight using the same structure as the fine aggregate calculation.
`def calculate_other_aggregate_AA(percent_other_I, sg_other_P, volume_X): return volume_X * (percent_other_I / 100) * sg_other_P * UNIT_WEIGHT_WATER`	This function calculates the weight of other aggregate materials in the mix.

`project_no = int(input("Enter project number: ")) concrete_class = input("Enter class of concrete: ")`	These lines collect general project information from the user. The project number is converted into an integer, while the concrete class is stored as a string.
`print("NDOT Concrete Mix Design - Weight Summary") print(f"Project Number: {project_no}") print(f"Concrete Class: {concrete_class}")`	These print statements display a formatted summary of the mix design results using f-strings for clean output formatting.
`print(f"Water Weight (Q): {Q:.2f} lb")`	This line prints the final water weight rounded to two decimal places using a floating-point format specifier.